求定积分

$$\int_0^{\pi} \frac{\pi + \cos x}{x^2 - \pi x + 2026} \mathrm{d}x$$

令$x - \frac{\pi}{2} = t$
得到

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\pi - \sin t}{t^2 - \frac{\pi^2}{4} + 2026} \mathrm{d}t$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \frac{\mathrm{d}t}{t^2 - \frac{\pi^2}{4} + 2026}$$

$$= \frac{2\pi}{\sqrt{2026 - \frac{\pi^2}{4}}} (\arctan \frac{\pi}{2\sqrt{2026 - \frac{\pi^2}{4}}})$$